

The aryl hydrocarbon receptor in inflammatory bowel disease: linking the environment to disease pathogenesis.



PURPOSE OF REVIEW: The aryl hydrocarbon receptor (AhR), a transcription factor activated by a large number of environmental agents, modulates the activity of immune and nonimmune cells in the gut, and may represent an important link between the environment and the immune perturbations which underlie the pathogenesis of inflammatory bowel disease. This review will summarize the current knowledge of the role of AhR in regulation of intestinal immune homeostasis and inflammation. RECENT FINDINGS: Activation of AhR by dietary ligands is necessary for the maintenance or expansion of innate immune cells in the gut, such as intraepithelial lymphocytes (IELs) and interleukin (IL)-22-producing lymphoid cells (ILC22). AhR-deficient mice lack IELs, have reduced number of ILC22 cells, and are more susceptible to bacterial infections and experimental colitis. In animal models, AhR activators inhibit proinflammatory cytokine synthesis and attenuate colitis by a pathway that involves IL-22. Analysis of AhR in the human gut reveals that intestinal T cells and natural killer cells isolated from Crohn's disease patients express low levels of AhR and respond to AhR ligands by downregulating inflammatory cytokines and upregulating IL-22. SUMMARY: These novel findings may help explain how environmental factors may regulate mucosal immune responses.